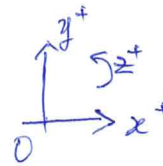
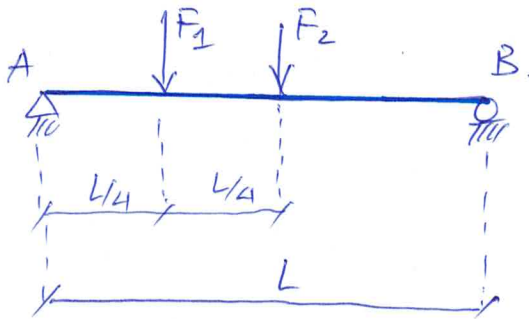


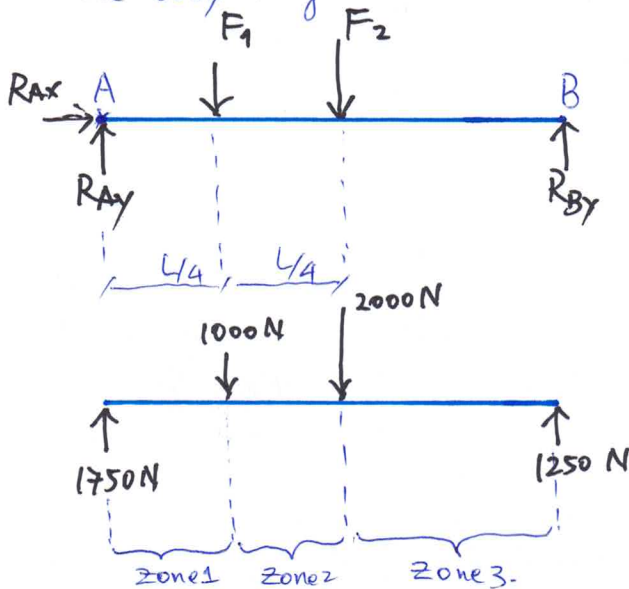
# Practice 4: Shear & Moment Diagrams

1.  $F_1 = 1000 \text{ N}$ ,  $F_2 = 2000 \text{ N}$ ,  $L = 10 \text{ m}$ .



Global CSys.

- Free Body Diagram



- Reaction Forces ?

$$\begin{cases} \sum F_x = 0 \Rightarrow R_{Ax} = 0 & (1) \\ \sum F_y = 0 \Rightarrow R_{Ay} + R_{By} - F_1 - F_2 = 0 & (2) \\ \sum M_{Any Point} = 0 \Rightarrow \sum M_A = 0 & (3) \end{cases}$$

$$(3) \Rightarrow +R_{By} \times L - F_1 \times \frac{L}{4} - F_2 \times \frac{L}{2} = 0$$

$$R_{By} = \frac{F_1}{4} + \frac{F_2}{2} = \frac{1000}{4} + \frac{2000}{2} = 1250$$

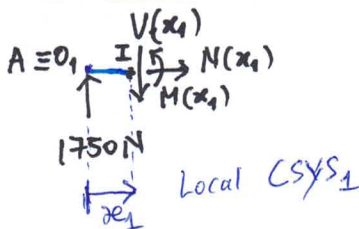
$$(2) \Rightarrow R_{Ay} = F_1 + F_2 - R_{By} = 1000 + 2000 - 1250$$

$$R_{Ay} = 1750 \text{ N}$$

- Zonal Forces : 3 zones.

- Cutting each zonal force at a point one by one and write equilibrium equations

⊕ Zone 1: Left side.



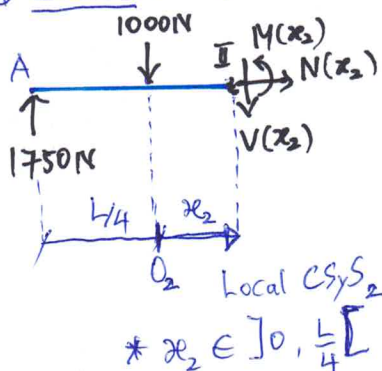
$$\sum F_x = 0 \Rightarrow N(x_1) = 0$$

$$\sum F_y = 0 \Rightarrow -V(x_1) + 1750 = 0 \Rightarrow V(x_1) = +1750 \text{ N}$$

$$\sum M_{z/I} = 0 \Rightarrow +M(x_1) - 1750 \times x_1 = 0 \Rightarrow M(x_1) = 1750 x_1 \text{ N.m}$$

$$* x_1 \in ]0, \frac{L}{4}[$$

⊕ Zone 2: Left Side.



$$\sum F_x = 0 \Rightarrow N(x_2) = 0$$

$$\sum F_y = 0 \Rightarrow +1750 - 1000 - V(x_2) = 0 \Rightarrow V(x_2) = -750 \text{ N}$$

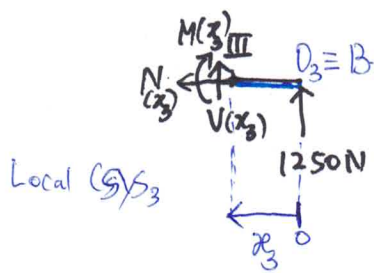
$$\sum M_{z/II} = 0 \Rightarrow +M(x_2) + 1000 \times x_2 - 1750 \times (\frac{L}{4} + x_2) = 0$$

$$\Rightarrow M(x_2) + 1000 x_2 - 1750 x_2 - 1750 \times \frac{10}{4} = 0$$

$$M(x_2) = 750 x_2 + 4375 \text{ [N.m]}$$

$$* x_2 \in ]0, \frac{L}{4}[$$

⊕ Zone 3: Right Side.



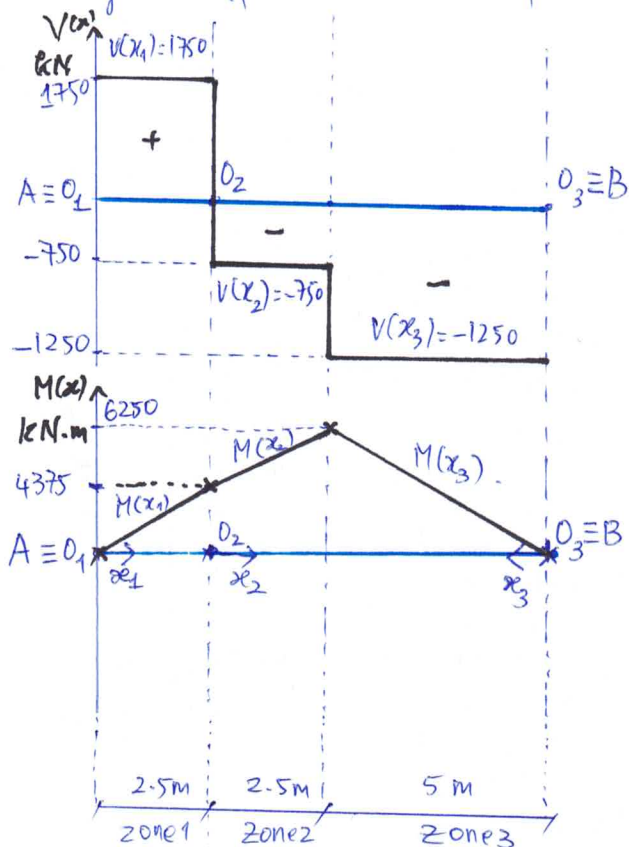
$$\sum F_x = 0 \Rightarrow +N(x_3) = 0$$

$$\sum F_y = 0 \Rightarrow +V(x_3) + 1250 = 0 \Rightarrow V(x_3) = -1250 \text{ N}$$

$$\sum M_{\frac{1}{3}} = 0 \Rightarrow +M(x_3) - 1250 \times x_3 = 0$$

$$M(x_3) = 1250x_3$$

- Diagram of  $V(x)$  &  $M(x)$  for all the 3 zones.



|          |   |      |
|----------|---|------|
| $x_1$    | 0 | 2.5  |
| $M(x_1)$ | 0 | 4375 |

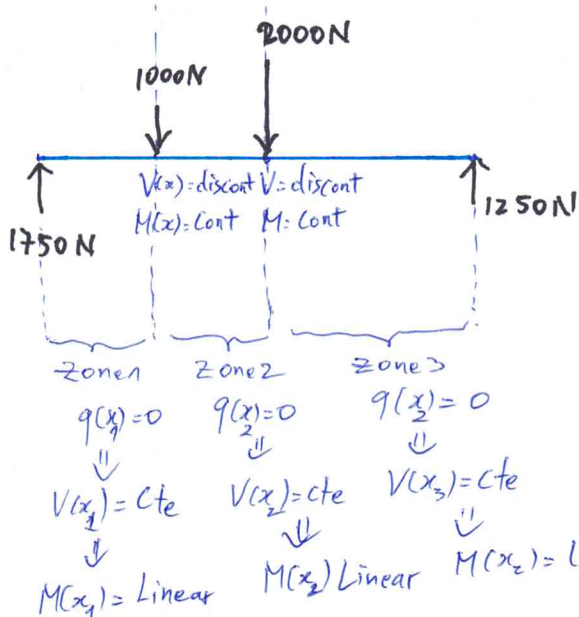
$$= 1750x_1$$

|          |      |      |
|----------|------|------|
| $x_2$    | 0    | 2.5  |
| $M(x_2)$ | 4375 | 6250 |

$$= 750x_2 + 4375$$

|          |   |        |
|----------|---|--------|
| $x_3$    | 0 | 2.5    |
| $M(x_3)$ | 0 | 1250x3 |

- Verification of  $V(x)$  &  $M(x)$  by  $q(x), V(x), M(x)$  relationship.



$$q(x) = -\frac{dV(x)}{dx}$$

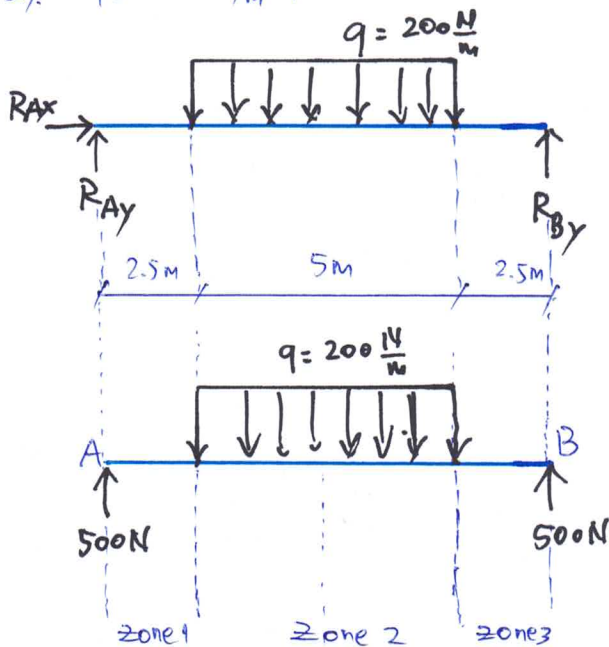
$$V(x) = \frac{dM(x)}{dx}$$

At border zone,

- between zone 1 & zone 2 =  $\left\{ \begin{array}{l} V: \text{discontinuous by } 1000 \text{ N} \\ M: \text{continuous} \end{array} \right.$

- between zone 2 & zone 3 =  $\left\{ \begin{array}{l} V: \text{discontinuous by } 2000 \text{ N} \\ M: \text{continuous} \end{array} \right.$

2).  $q = 200 \text{ N/m}$ ,  $L = 10 \text{ m}$ .



$$\sum F_x = 0 \Rightarrow R_{Ax} = 0$$

$$\sum F_y = 0 \Rightarrow R_{Ay} + R_{By} - (200 \times 5) = 0.$$

$$\sum M_A = 0 \Rightarrow R_{By} \times 10 - (200 \times 5) \times 5 = 0.$$

$$\Rightarrow R_{By} = \frac{200 \times 5 \times 5}{10} = 500 \text{ N}$$

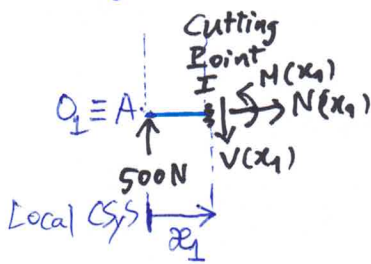
$$\Rightarrow R_{Ay} = (200 \times 5) - R_{By}$$

$$R_{Ay} = 1000 - 500 = 500 \text{ N}$$

← Symmetric →  
Line.

$V(x)$  is anti-symmetric about symmetric line.  
 $M(x)$  = Symmetric

⊕ Zone 1: Left side.



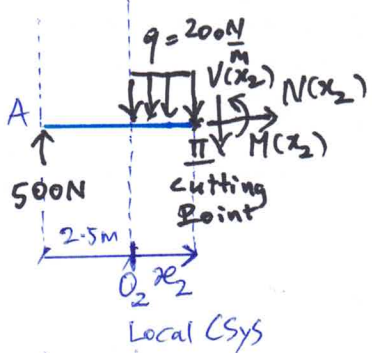
$$\sum F_x = 0 \Rightarrow N(x_1) = 0$$

$$\sum F_y = 0 \Rightarrow -V(x_1) + 500 = 0 \Rightarrow V(x_1) = +500 \text{ N}$$

$$\sum M_{x1} = 0 \Rightarrow M(x_1) - 500 \times x_1 = 0 \Rightarrow M(x_1) = +500x_1$$

|          |   |     |
|----------|---|-----|
| $x_1$    | 0 | 2.5 |
| $M(x_1)$ | 0 | 750 |

⊕ Zone 2: Left side.



$$\sum F_x = 0 \Rightarrow N(x_2) = 0$$

$$\sum F_y = 0 \Rightarrow -V(x_2) + 500 - (200 \times x_2) = 0 \Rightarrow V(x_2) = -200x_2 + 500$$

$$\sum M_{x2} = 0 \Rightarrow M(x_2) + (200 \times x_2) \times \frac{x_2}{2} - 500(x_2 + 2.5) = 0$$

$$\Rightarrow M(x_2) = -100x_2^2 + 500x_2 + 750$$

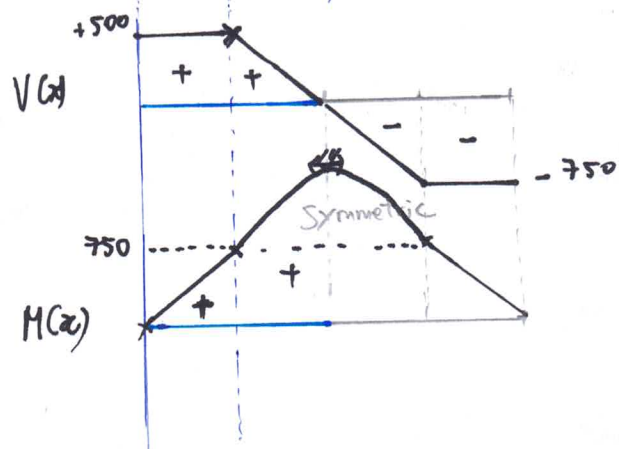
$$M'(x_2) = -200x_2 + 500$$

$$M'(x_2) = 0 \Rightarrow -200x_2 + 500 = 0 \Rightarrow x_2 = \frac{500}{200} = 2.5 \text{ m}$$

Extremum Point

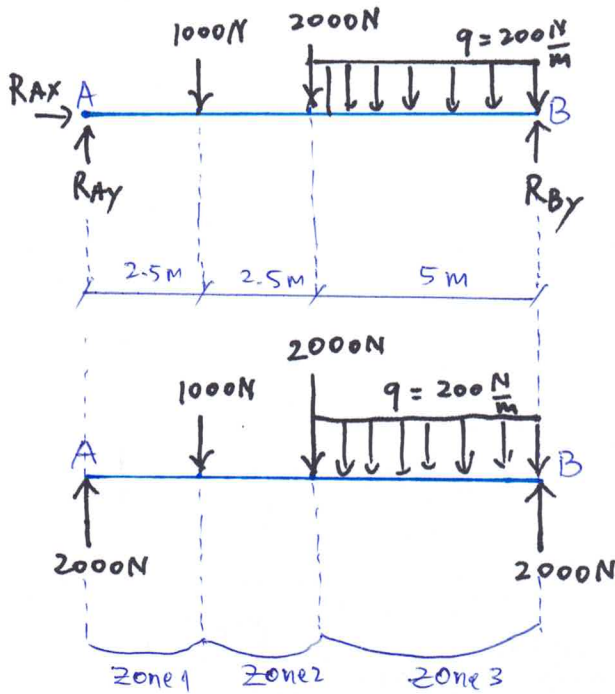
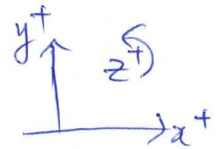
|          |     |      |
|----------|-----|------|
| $x_2$    | 0   | 2.5  |
| $M(x_2)$ | 750 | 1375 |

Symmetric Point.





3.  $F_1 = 1000 \text{ N}$ ;  $F_2 = 2000 \text{ N}$ ,  $q = 200 \frac{\text{N}}{\text{m}}$ ,  $L = 10 \text{ m}$ .



$$\sum F_x = 0 \Rightarrow R_{Ax} = 0$$

$$\sum F_y = 0 \Rightarrow R_{Ay} + R_{By} - 1000 - 2000 - (200 \times 5) = 0$$

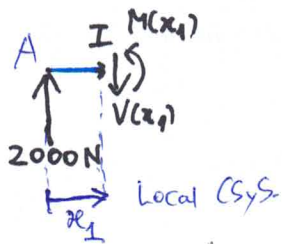
$$R_{Ay} + R_{By} = 4000$$

$$\sum M_{z/A} = 0 \Rightarrow R_{By} \times 10 - 1000 \times 2.5 - 2000 \times 5 - (200 \times 5) \times 7.5 = 0$$

$$\Rightarrow R_{By} = \frac{2500 + 10000 + 7500}{10} = 2000 \text{ N}$$

$$\Rightarrow R_{Ay} = 4000 - 2000 = 2000 \text{ N}$$

⊕ Zone 1: Left Side.



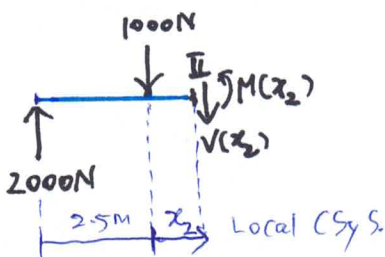
$$\sum F_x = 0 \Rightarrow N(x_1) = 0$$

$$\sum F_y = 0 \Rightarrow -V(x_1) + 2000 = 0 \Rightarrow V(x_1) = +2000 \text{ N}$$

$$\sum M_I = 0 \Rightarrow M(x_1) - 2000x_1 = 0 \Rightarrow M(x_1) = +2000x_1 \text{ N.m}$$

|          |   |      |
|----------|---|------|
| $x_1$    | 0 | 2.5  |
| $M(x_1)$ | 0 | 5000 |

⊕ Zone 2: Left side.



$$\sum F_y = 0 \Rightarrow -V(x_2) - 1000 + 2000 = 0 \Rightarrow V(x_2) = 1000 \text{ N}$$

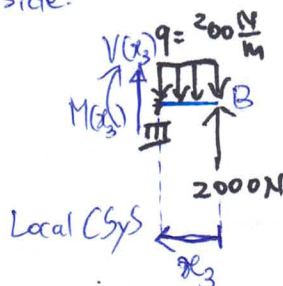
$$\sum M_{II} = 0 \Rightarrow M(x_2) + 1000x_2 - 2000(2.5 + x_2) = 0$$

$$M(x_2) = -1000x_2 + 2000x_2 + 5000$$

$$M(x_2) = 1000x_2 + 5000$$

|          |      |      |
|----------|------|------|
| $x_2$    | 0    | 2.5  |
| $M(x_2)$ | 5000 | 7500 |

⊕ Zone 3: Right Side.



$$\sum F_y = V(x_3) + 2000 - (200 \times x_3) = 0$$

$$V(x_3) = 200x_3 - 2000$$

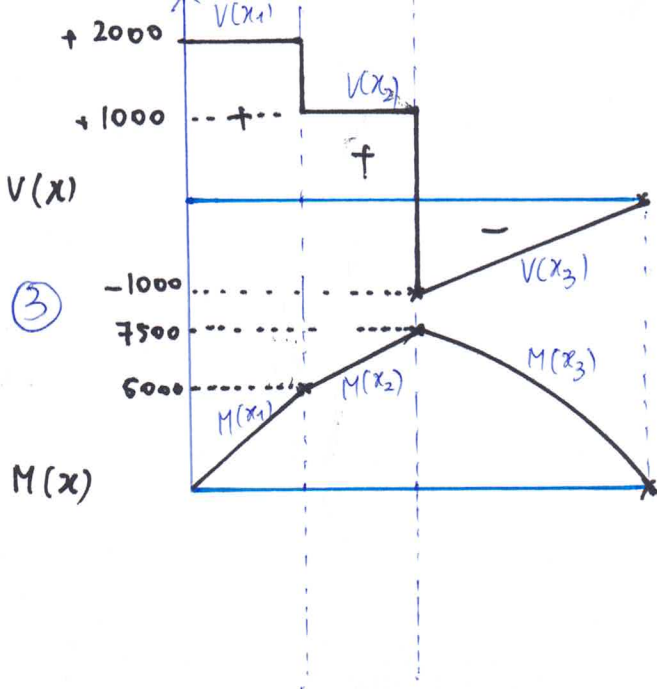
$$\sum M_{z/III} = 0 \Rightarrow M(x_3) - 2000x_3 + (200x_3) \times \frac{x_3}{2} = 0$$

$$M(x_3) = -100x_3^2 + 2000x_3$$

$$M'(x_3) = -200x_3 + 2000 = 0$$

$$\Rightarrow x_3 = 10 \text{ Extremum Point}$$

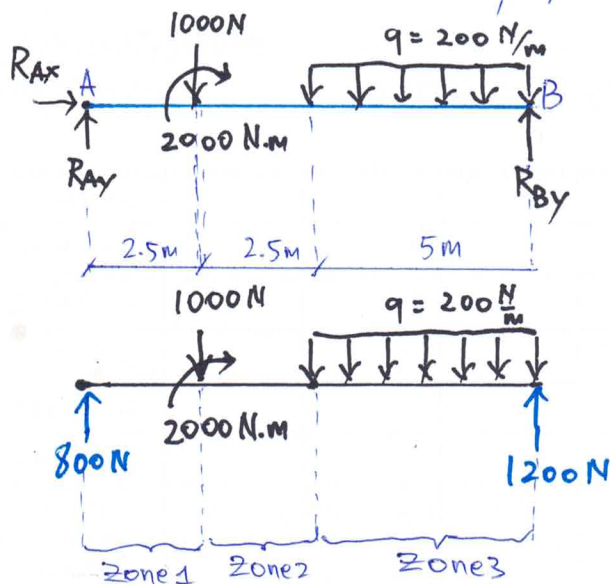
|      |   |          |
|------|---|----------|
| 5m   | 0 | $x_3$    |
| 7500 | 0 | $M(x_3)$ |



$V(x)$  is discontinuous at both border points because of external vertical forces:  $F_1 \times F_2$

$M(x)$  is ~~dis~~continuous at both border points because of no external bending moment

④ -  $F = 1000\text{ N}$ ;  $M_e = 2000\text{ N.m}$ ;  $q = 200\text{ N/m}$ ;  $L = 10\text{ m}$ .



$$\sum F_x = 0 \Rightarrow R_{Ax} = 0$$

$$\sum F_y = 0 \Rightarrow R_{Ay} + R_{By} - 1000 - (200 \times 5) = 0$$

$$R_{Ay} + R_{By} = 2000$$

$$\sum M_{z/A} = 0 \Rightarrow R_{By} \times 10 - (200 \times 5) \times 7.5 - 1000 \times 2.5 - 2000 = 0$$

$$R_{By} = \frac{7500 + 2500 + 2000}{10} = 1200\text{ N}$$

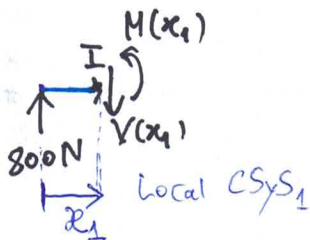
$$\Rightarrow R_{Ay} = 2000 - R_{By} = 800\text{ N}$$

⊕ zone 1: Left Side.

$$\sum F_y = 0 \Rightarrow -V(x_1) + 800 = 0 \Rightarrow V(x_1) = 800\text{ N}$$

$$\sum M_{z/I} = 0 \Rightarrow M(x_1) - 800 \times x_1 = 0 \Rightarrow M(x_1) = 800 x_1 [\text{N.m}]$$

|       |          |   |      |
|-------|----------|---|------|
| [m]   | $x_1$    | 0 | 2.5  |
| [N.m] | $M(x_1)$ | 0 | 2000 |



⊕ zone 2: Left Side.

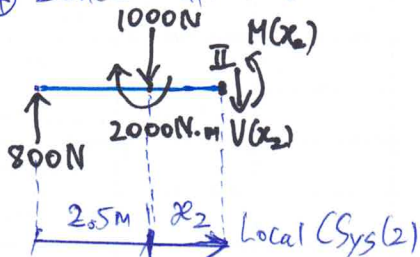
$$\sum F_y = 0 \Rightarrow -V(x_2) - 1000 + 800 = 0 \Rightarrow V(x_2) = -200\text{ N}$$

$$\sum M_{z/II} = 0 \Rightarrow M(x_2) - 2000 + 1000 \times x_2 - 800(2.5 + x_2) = 0$$

$$M(x_2) = -1000x_2 + 800x_2 + 2000 + 800 \times 2.5$$

$$M(x_2) = -200x_2 + 4000$$

|          |      |      |
|----------|------|------|
| $x_2$    | 0    | 2.5  |
| $M(x_2)$ | 4000 | 3500 |



⊕ zone 3: Right Side.

$$\sum F_y = 0 \Rightarrow V(x_3) + 1200 - (200 \times x_3) = 0$$

$$V(x_3) = 200x_3 - 1200$$

|     |          |      |       |
|-----|----------|------|-------|
| [m] | $x_3$    | 5    | 0     |
| [N] | $V(x_3)$ | -200 | -1200 |

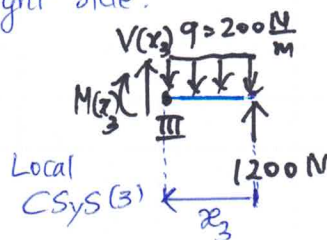
$$\sum M_{z/III} = 0 \Rightarrow M(x_3) + (200x_3) \times \frac{x_3}{2} - 1200x_3 = 0$$

$$M(x_3) = -100x_3^2 + 1200x_3$$

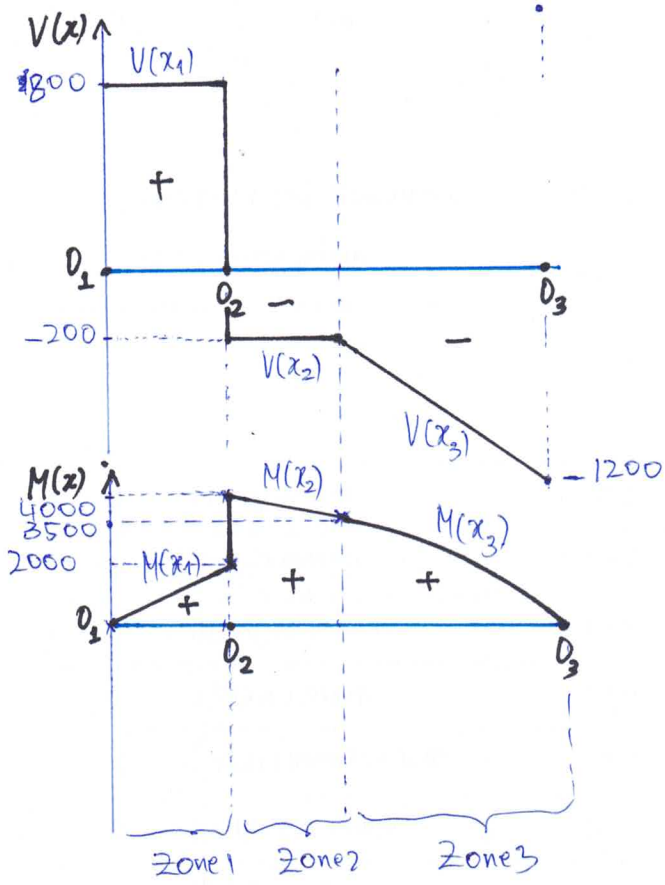
$$M'(x_3) = -200x_3 + 1200$$

Extremum:  $M'(x_3) = 0 \Rightarrow x_3 = \frac{1200}{200} = 6\text{ m}$   
Point:

|       |          |      |   |
|-------|----------|------|---|
| [m]   | $x_3$    | 5    | 0 |
| [N.m] | $M(x_3)$ | 3500 | 0 |



④



\* At border zones :

- Between zone 1 & zone 2.  
 $V(x)$  discontinuous coz of 1000N  
 $M(x)$  discontinuous coz of  $M_e = 2000$
- Between zone 2 & zone 3  
 $V(x)$  is continuous since there are  
 $M(x)$  is continuous external force  
 at that  
 border point